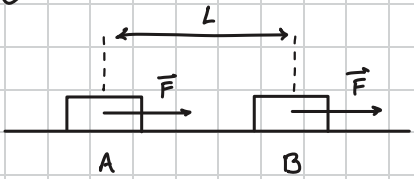
Week 6

6.1 Work

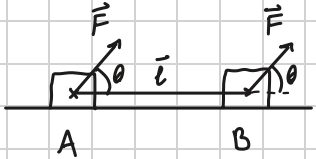
①



Constant force same direction as displacement

$W_{A \to B} = F \cdot L$, $\langle w \rangle = N \cdot m = J$

②



$\vec{l}$ displacement vector
constant force at an angle θ

$W_{A \to B} = \vec{F} \cdot \vec{l} = F l \cos\theta$

Note : Work can't be negative

if $\frac{\pi}{2} < \theta \leq \pi \Rightarrow \cos\theta < 0 \Rightarrow w < 0$

Example : friction forces do negative work

Note : if $\vec{F} \perp \vec{l} \Rightarrow \theta = \frac{\pi}{2} \Rightarrow w = 0$

Example : Centripetal forces do no work

Several forces acting on an object

$$\vec{F}_1, \vec{F}_2, \vec{F}_3 \ldots \vec{F}_n$$

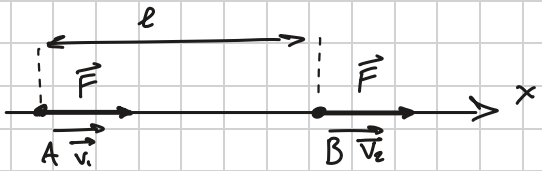
Resultant force $\vec{F} = \sum_{k=1}^{n} \vec{F}_k$

$$W_{TOT} = \vec{F} \cdot \vec{l} = \left(\sum_{k=1}^{n} \vec{F}_k \right) \cdot \vec{l}$$

$$= \underbrace{\vec{F}_1 \cdot \vec{l}}_{W_1} + \underbrace{\vec{F}_2 \cdot \vec{l}}_{W_2} + \ldots \underbrace{\vec{F}_n \cdot \vec{l}}_{W_n}$$

$$\boxed{W_{TOT} = W_1 + W_2 + \cdots W_n}$$

Work energy



$$F = m \cdot \vec{a} \quad \Rightarrow \quad F = ma \quad \Rightarrow \quad a = \frac{F}{m}$$

a is constant

$$V_2^2 - V_1^2 = 2ad$$

$$\Rightarrow \quad a = \frac{V_2^2 - V_1^2}{2d}$$

$$\Rightarrow \quad \left(\frac{F}{m} = \frac{V_2^2 - V_1^2}{2d} \right) \cdot md$$

$$F \cdot d = \frac{m(V_2^2 - V_1^2)}{2} \quad \text{or} \quad F \cdot d = \frac{mV_2^2}{2} - \frac{mV_1^2}{2}$$

$KE = \frac{mv^2}{2}$ kinetic energy

$\langle KE \rangle = kg \frac{m^2}{s^2} = kg \cdot \frac{m}{s^2} \cdot m = N \cdot m = J$

$$W_{TOTAL} = KE_2 - KE_1$$

Work-Energy theorem

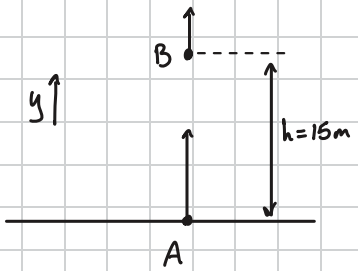
Example: in circular motion with constant speed, $\vec{F}_{cp} \perp$ displacement

$W_{tot} = 0 = \Delta KE \Rightarrow KE = \frac{mv^2}{2} = \text{Constant}$

Ex rock w 3.0N thrown up ↑

when it reaches 15m, $v = 25 m/s$

a) find V_i



$\vec{a} = g$

$W_{g\,A\to B} = -mgh < 0$

$$W_{g\,A\to B} = \frac{mv^2}{2} = \frac{mV_0^2}{2}$$

$$\left(-mgh = \frac{mv^2}{2} - \frac{mV_0^2}{2}\right) \cdot \frac{1}{m} \cdot 2$$

$$-2gh = v^2 - V_0^2 \Rightarrow \boxed{V_0^2 = v^2 + 2gh}$$

$$V_0 = \sqrt{v^2 + 2gh}$$

$$= 30.3 m/s$$

a) max height?

At max $v = 0$ (height = c)

$$W_{A\to C} = -mgh_{max}$$

$$W_{A\to C} = \frac{mv^{2\,0}}{2} - \frac{mV_0^2}{2}$$

$$\Rightarrow \left(-mgh_{max} = -\frac{mV_0^2}{2}\right) \cdot \left(-\frac{2}{m}\right)$$

$$2gh_{max} = V_0^2$$

$$\boxed{h_{max} = \frac{V_0^2}{2g}}$$

$$= 46.9 m$$